Milestone 2: Early Warning and Decision Support for Animal Shelter Overcrowding

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1. Project Context

Animal shelters must manage a continuous flow of animals entering and leaving while operating with limited space, staff, and other resources. Intake and outflow can change over time, which means a shelter that is operating normally today may face capacity pressure in the near future. If overcrowding is recognized only after capacity becomes critical, managers may have less time and fewer options to respond.

This project investigates whether shelter records can support earlier assessment of near-term capacity pressure. The goal is to give shelter managers enough information and time to consider actions such as transfers, foster placement, or adoption efforts before overcrowding becomes critical.

The project focuses on short-term, shelter-level capacity management. It does not attempt to manage all shelter operations or make decisions about individual animals. The analytical system is intended to support shelter managers, while final operational decisions remain with them.

2. Decision or Scientific Question

The central decision question is:

> **Based on information available today, is the shelter at meaningful risk of overcrowding in the near future, and should the manager begin considering a response?**

This question is evaluated at the shelter level at regular decision times as new information becomes available. The proposed system would use available shelter conditions to assess near-term capacity pressure once historical occupancy and capacity measures can be verified. A future operational output should report capacity risk and its uncertainty rather than only a yes-or-no prediction. This information can help managers decide whether to continue normal operations, monitor the situation more closely, or consider intervention. For the preliminary analysis, the manager receives a forecast after the end-of-day records for date t are finalized, with a seven-day horizon covering dates $t+1$ through $t+7$. The long-term decision target remains capacity risk; the presently measurable auxiliary target is future net event flow, not verified overcrowding.

3. Stakeholder, Decision Maker, and Intended Value

The primary users are shelter managers and operations staff responsible for monitoring capacity and coordinating responses. They must manage changing animal intake and outflow while working with limited space, staff, foster capacity, and other resources.

The system is intended to give these decision makers earlier and more structured information about possible capacity pressure. Earlier warning may provide more time to review the situation and prepare a response before overcrowding becomes critical. The system does not replace managerial judgment: managers remain responsible for the final decision and can consider operational information that may not be captured in the data.

4. Alternative Actions and Expected Outcomes

Based on the estimated risk and current shelter conditions, managers may continue normal operations, monitor the situation more closely, or prepare a response. Possible responses include transfers, foster placement, and adoption efforts, depending on the resources available at that time.

The main tradeoff is between acting too late and responding unnecessarily. A useful early warning may give managers more time to prevent or reduce capacity pressure, while a missed warning may leave fewer options once overcrowding occurs. A false warning may also waste staff time and limited resources. When the estimate is highly uncertain, additional monitoring or human review may be more appropriate. If the system is unavailable, managers will continue using the existing operating procedures.

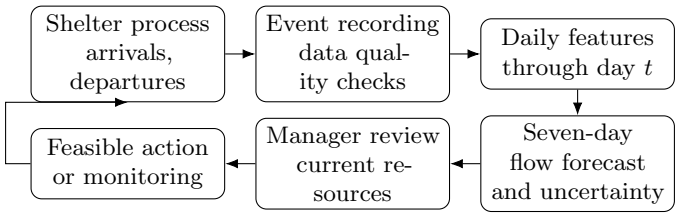
5. Information Available at Decision Time

At each decision time t, the system will use only information that is already available to the shelter. The exact variables will depend on the final dataset, but the main information groups are summarized below.

Information Type	Examples	Use
Available at decision time	Current occupancy, recent intake and outflow, shelter capacity, animal composition, date and seasonal information	Inputs to the system
Generated by the system	Estimated future capacity pressure, overcrowding risk, and uncertainty	Information for the manager
Available only afterward	Future intake and outflow, realized occupancy, and whether overcrowding occurred	Evaluation and monitoring
Useful but availability uncertain	Foster availability, partner-shelter capacity, staffing, and scheduled events	Additional context when available

Information that becomes known only after time t will not be used to make an earlier prediction. This time boundary is necessary to prevent information leakage and to make the evaluation reflect how the system would operate in practice.

6. Updated Connected Decision Architecture



Interpretation. Actions can change later shelter conditions, which then produce new records. The flow model is predictive, not an estimate of intervention effects. True capacity warnings additionally require verified onsite inventory and matched capacity data.

7. Decision-System Canvas

The following canvas updates the initial project formulation based on the current decision-system definition. It connects the shelter process, decision-time information, prediction, uncertainty, alternative actions, managerial decisions, and observed outcomes. Components that depend on the final dataset or operational setting remain preliminary or unresolved.

Decision Objective. Use information available at the current decision time to estimate near-term shelter capacity pressure and provide shelter managers with enough warning to consider an appropriate response before overcrowding becomes critical. The system provides decision support rather than automatically selecting or implementing an intervention.

Component	Current Formulation	Status / Notation
1. **Decision / scientific objective**	Estimate whether the shelter faces meaningful near-term capacity pressure and support the manager in deciding whether to continue normal operations, increase monitoring, or begin considering a response.	✓
2. **Stakeholder / intended user / value**	Shelter managers and operations staff responsible for monitoring capacity and coordinating responses. Earlier warning may provide more time to prepare before overcrowding becomes critical. Final decisions remain with managers.	✓
3. **Decision or inference time**	The system will be evaluated at regular decision times as new shelter information becomes available. The preliminary forecast is issued after each day's records are finalized, using a seven-day prediction horizon.	?
4. **Information available at that time**	Recent recorded intake/outcome counts and calendar information. Current onsite occupancy, historical capacity, staffing, and foster counts are not assumed observed in the current research dataset. Information observed only afterward will not be used for earlier predictions.	X_t
5. **Data-generating system**	Shelter occupancy changes as animals enter and leave over time. Intake, adoption, transfer, foster placement, return-to-owner, and other processes may vary across time and seasons, creating temporal dependence in the data.	△
6. **Uncertain target / state**	Observed preliminary target: seven-day cumulative net event flow. Intended future target: onsite occupancy relative to matched historical capacity; this cannot yet be computed from validated daily measurements.	Y_t, ?
7. **Learning requirement**	Learn whether information available at time t contains useful information about future capacity pressure while preserving the true time order of the shelter process and avoiding information leakage.	f_θ
8. **Prediction / inference output**	Currently, predicted seven-day net flow and error summary; a validated capacity-risk estimate and uncertainty interval remain proposed.	$\widehat{Y}_t^{(7)}$, △
9. **Uncertainty**	Future intake and outflow are uncertain. Prediction quality may also be affected by seasonal variation, unusual intake events, estimation error, and changes in shelter operations.	u_t
10. **Feasible alternatives**	Managers may continue normal operations, monitor the situation more closely, or prepare a response such as transfer, foster placement, or additional adoption efforts when operationally feasible.	$a \in \mathcal{A}_t$
11. **Objective / criterion**	The decision should balance the consequences of acting too late against the cost and resource use of responding unnecessarily. Prediction accuracy alone is therefore not the final objective of the system.	ℓ_D
12. **Decision rule / intended use**	Risk estimates and uncertainty are provided to the shelter manager as decision-support information. The manager combines this information with current resources and operational context before deciding whether and how to respond.	△
13. **Action-dependent outcome?**	Yes. Actions such as transfers, foster placement, and adoption efforts may change future occupancy. However, the predictive model alone will not be used to estimate or claim the causal effect of these interventions.	$Y_t(a)$
14. **Major assumptions**	The records provide useful event-flow measurements but do not directly measure onsite occupancy or capacity. The analysis assumes past event patterns may help forecast future flow; whether a warning provides enough lead time for action remains to be evaluated.	△
15. **Highest technical risks**	Major risks include incomplete or inconsistent data, an unclear definition of overcrowding, temporal dependence or information leakage, changes in shelter operations over time, and predictions that do not provide enough lead time for useful action.	?

Status notation: ✓ = established in the current project definition; △ = preliminary and will be tested; ? = unresolved and will be refined in later milestones.

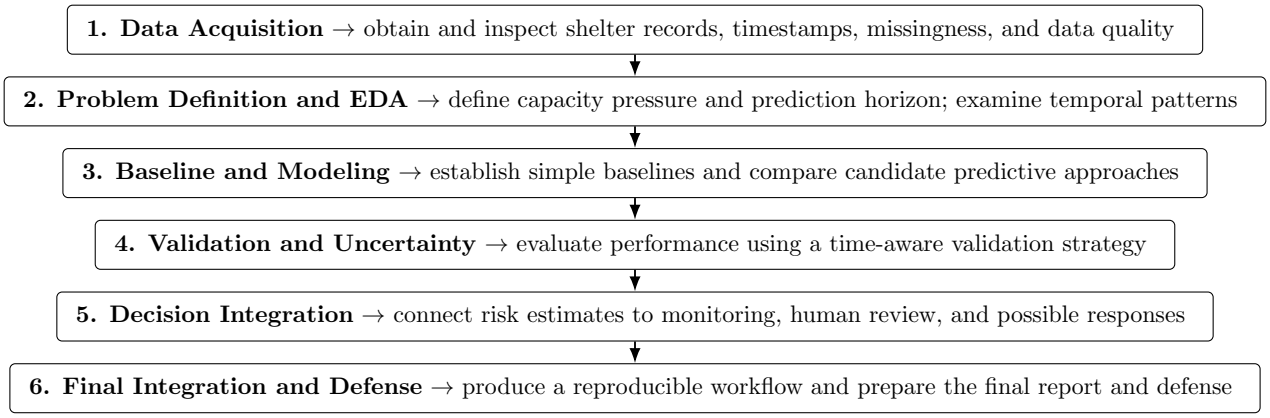
8. Initial Assumptions

The preliminary analysis assumes that recorded intake and outcome events are sufficiently consistent to study historical daily flow. It does not assume that these records alone reconstruct onsite occupancy: animals in foster care, changes in reporting practices, and missing historical capacity measures complicate that inference. Recent flow and calendar patterns may contain predictive information, but their stability across operational periods needs testing.

For the proposed decision system, an alert must also arrive early enough for managers to act. The seven-day horizon is an initial design choice, not yet validated against transfer, foster-placement, or staffing lead times.

9. Project Work Plan and Milestones

The project follows six stages. Data inspection, preliminary target construction, and simple baseline evaluation have begun; operational risk prediction and decision evaluation remain future work.



The results of each stage may change the design of later stages. Documentation, reproducibility, and updates to the Decision-System Canvas will continue throughout the project.

10. Major Technical Risks

Data availability and measurement. Shelter records may be incomplete or may not provide a clear measure of capacity pressure. We will examine data coverage, missingness, and the definitions of occupancy and capacity. If necessary, the project can use a more reliable subset of the data or redefine the analytical target.

Temporal structure. Shelter conditions are observed over time, so observations may not be independent. Using future information or random validation could therefore produce unrealistic performance estimates. The project will preserve the true time order when constructing features and evaluating the system.

Changes over time. Intake patterns, capacity, or shelter operations may change, making historical relationships less useful for future prediction. We will compare patterns and performance across time and revise or limit the model if these relationships are not stable.

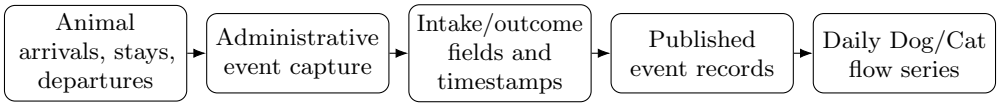
Decision usefulness. An accurate prediction may still have little value if it comes too late for managers to respond. The selected seven-day prediction horizon must still be compared with realistic response times and available actions. If direct intervention cannot be supported, the system may instead be used for earlier monitoring and human review.

Finally, the predictive model will not be used to claim that a particular intervention causes a better outcome. Determining the effects of transfers, foster placement, or adoption efforts would require additional evidence beyond this predictive analysis. The project will therefore keep prediction and intervention effects separate.

11. Population, Data-Generating Process, and Provenance

System and scope. The source system is Austin Animal Center and its recorded animal intake/outcome operations. The present sample is a historical extract ending in early May 2025; the analysis is not a multi-shelter sample and cannot establish performance for all US shelters. The intended near-term deployment population is future days at the same operational setting, subject to changes in policy, reporting, and capacity.

Data-generating chain.



Records are not random draws from all animals or all shelters; they arise from a particular administrative process. Repeat admissions, coding rules, late entry, time-zone conventions, onsite versus offsite foster status, and outcome classification may distort a direct mapping from event flow to physical kennel occupancy. Daily zeros may reflect either no activity or missing reporting; completeness needs further examination.

Provenance. Public City of Austin / Austin Animal Center historical Intakes and Outcomes data, downloaded as CSV and retained unchanged in local `data/raw/`. Dataset identifiers: intakes (wter-evkm) and outcomes (9t4d-g238). The particular local extracts contain 173,812 intake events and 173,775 outcome events (12 columns each), spanning approximately October 2013 to May 2025. The exact extraction timestamp was not captured in the available analysis records; this limits version-level reproducibility for a changing public dataset. All derived counts are computed from the two immutable source CSVs.

12. Observational Units and Mathematical Contract

There are three relevant levels: (i) an *event* row; (ii) an animal identified by Animal ID that can appear multiple times; and (iii) a *shelter-day* t, the observational unit for the preliminary forecast. Daily observations are ordered and potentially dependent; repeated records from one animal are not independent stays.

Let I_t and E_t be the numbers of recorded Dog/Cat intake and outcome events on calendar day t (units: events/day), and $F_t = I_t - E_t$ the net recorded flow. Define the post-close information set $\mathcal{H}_t$ as all records known after day t has been finalized. For the preliminary seven-day forecast,

$$Y_t^{(7)} = \sum_{h=1}^{7} F_{t+h}, \qquad \widehat{Y}_t^{(7)} = f_\theta(X_t, C_t), \tag{1}$$

where $X_t \in \mathbb{R}^p$ contains lagged intake/outcome features from $\mathcal{H}_t$, C_t contains calendar context (e.g., month, weekday), and $Y_t^{(7)} \in \mathbb{Z}$ is the *future cumulative net event flow* (units: events per seven-day window). No future realized event appears in X_t. Window length $H = 7$, lag selection, species aggregation, time period, and any eventual model tuning are structural choices to be assessed.

The eventual decision quantity differs: with measured onsite population $O_{s,t}^{\text{onsite}}$ and corresponding available capacity $K_{s,t}$ for species/resource group s, a prospective capacity ratio would be $R_{s,t} = O_{s,t}^{\text{onsite}}/K_{s,t}$, and a prospective overcrowding event could be $\max_{1\le h\le 7} R_{s,t+h} > 1$ (threshold provisional). Neither historic daily O^{onsite} nor contemporaneous K has yet been validated. Thus $Y_t^{(7)}$ is *not* labeled actual overcrowding.

13. Measurement and Data Quality

Variable	Meaning / construction	Status and limitation
Animal ID	Administrative animal identifier (not a unique stay)	Repeated across episodes
Event timestamp	Intake parsed from US date-time text; outcome day taken from recorded ISO calendar date	Mixed aware/naive outcome formats; timezone semantics unresolved
Animal type	Dog/Cat subset for flow analysis	Other categories excluded explicitly
F_t	Count(intakes) minus count(outcomes) for day t	Derived event-flow measure
$Y_t^{(7)}$	Sum of flows on days $t+1$ through $t+7$	Derived, only where all seven days exist
Onsite population, capacity	Animals physically housed and resource-matched available space	Not yet verified historical daily data

The two datasets have 156,287 and 156,237 distinct intake and outcome animal IDs respectively; both have zero missing Animal IDs. Across the union of 157,101 IDs, 155,215 (98.80%) have equal numbers of intake and outcome events, but equal event counts do not prove correctly reconstructed stays. Tentative chronological event-pairing produces 781 negative and 19 zero candidate durations; 642 negative cases are same-calendar-day pairs, including many midnight-coded records. Such intervals are diagnostics, not credible negative stay lengths and are not silently repaired or used as true occupancy.

During daily aggregation, the calendar index was completed and dates without recorded intake or outcome events were assigned zero counts. This treats absent event dates as days with no recorded activity rather than unreported activity; that assumption has not been independently verified. It is a potential source of measurement error, especially around gaps or reporting changes.

Outcome timestamps mix 170,191 timezone-naive values and 3,584 values with explicit offsets; therefore preliminary daily counts use the recorded calendar date without asserting all naive times are UTC. Missingness by column is emitted by the notebook, but no mechanism of missingness is inferred from counts alone. The Outcome Type field has 46 missing values; rare classes and large daily counts are retained pending source inspection.

As a limited external check, September 2023 source CSVs have 926 intakes and 873 outcomes across all animal types (net +53), while an official monthly inventory comparison suggests total-in-care increased by 47 and onsite count increased by 104. Those are different populations/measurement conventions; a single approximate agreement in total-in-care is *not* validation of daily onsite occupancy. Reported official monthly Dog/Cat event counts also show small disagreements with the CSV extraction. Reporting definitions, snapshot timing, and subsequent data corrections are possible explanations, not confirmed causes.

14. Initial EDA and Dependence Evidence

The derived Dog/Cat shelter-day table contains 4,235 dates (2013-10-01 to 2025-05-05), with 163,932 recorded intakes and 163,904 outcomes. Daily event counts are not interchangeable with contemporaneous animal inventory.

The early period exhibits warmer-month activity peaks. Long-run plots suggest lower overall flows after 2020, but the cause is unknown. Raw daily intake autocorrelation from 2014–2019 is $\rho(1) = 0.366$, $\rho(7) = 0.342$, and $\rho(14) = 0.341$, demonstrating that a row-level IID assumption is unsuitable. After subtraction of pooled

2014–2019 month	Intake/day	Outcome/day	Net/day
January	37.39	39.44	−2.05
May	60.91	50.98	+9.93
June	59.46	57.40	+2.06
July	52.17	56.81	−4.64
December	37.17	42.85	−5.68

Table 1: Selected monthly averages of daily Dog/Cat recorded events, 2014–2019. The May imbalance motivates further pressure analysis but does not show actual capacity exceedance.

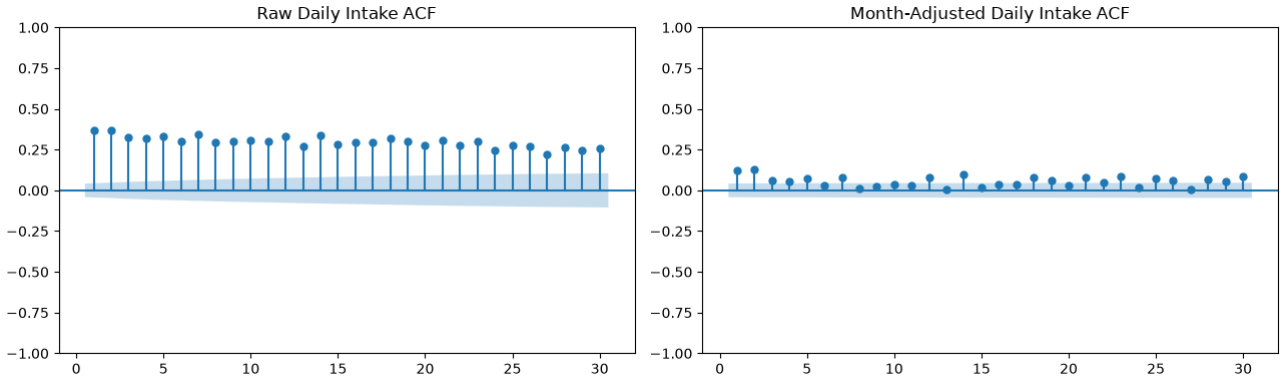


Figure 1: Autocorrelation of recorded daily Dog/Cat intakes (2014–2019), before and after subtracting the pooled mean for each calendar month. The adjustment reduces, but does not eliminate, the observed time dependence. Source: analysis notebook, Step 15.

calendar-month means (Figure 1), lag-14 autocorrelation falls to approximately 0.096; monthly seasonal patterns appear to account for part of the observed temporal structure, although residual dependence and changes across years remain. Month-adjustment here is exploratory; fitted seasonal statistics in an operational model must be learned from past/training data only.

The seven-day target has 4,228 fully observed labels: mean 0.014, median 0, standard deviation 50.88, range $[-237, 217]$, and 2,070 positive versus 2,110 negative observations (48 zero). Overlapping windows induce additional dependence. Target standard deviations vary between 2014–2019 (58.32), 2020 (34.61), and 2021–2024 (43.16); this suggests potential nonstationarity, not a proven causal structural break. The accompanying notebook provides the calculations and plots underlying these descriptive results. The data snapshots and instructions needed to reproduce the analysis are available in the GitHub repository.

15. Leakage Risks and Intended Generalization

The intended evaluation concerns *later dates at the same shelter*, not randomly held-out event rows, unseen animal IDs, other shelters, or guaranteed future years. For a forecast issued at the end of day t, allowable inputs must be in $\mathcal{H}_t$. The seven-day label may use future outcomes solely for historical supervision/evaluation.

Preliminary split. Development dates 2014-01-01 to 2023-12-24 ($n = 3,645$); validation forecast dates 2024-01-01 to 2024-12-24 ($n = 359$). The excluded boundary period prevents development labels from including 2024 validation outcomes. As a future refinement, choose model and hyperparameters using time-ordered internal folds; reserve a later untouched period as final test, with a seven-day outcome-availability rule at each training cutoff. Since the present baselines do not fit parameters, this is an out-of-time

Risk	Control / interpretation
Future target in predictors	Create target with leads, features only with present/past events
Overlapping seven-day labels	Purge development labels touching holdout dates; do not treat 359 validation dates as independent replicates
Random shuffling of temporal observations	Use chronological holdout; propose rolling-origin evaluation
Preprocessing or monthly means on all periods	Fit any data-driven transforms on development fold only
Animal repeat episodes and date mismatches	Never infer independent stay durations from ordinal ID pairing alone
Regime and onsite/offsite differences	Report temporal coverage and test future periods separately; do not label flow as occupancy

descriptive comparison rather than a tuned-model test.

16. Preliminary Baselines and Analytical Architecture

Two predetermined benchmarks predict future seven-day event flow:

$$\widehat{Y}_t^{\text{zero}} = 0, \qquad \widehat{Y}_t^{\text{recent7}} = \sum_{h=0}^{6} F_{t-h}. \tag{2}$$

On the 2024 chronological holdout, results are:

Baseline	MAE (events/7 days)	RMSE (events/7 days)
Zero flow	34.774	46.698
Recent seven-day flow	33.000	42.694

The recent-flow baseline reduces MAE by approximately 5.1% and RMSE by 8.6% relative to zero; the gain is modest, not a significance claim or an overcrowding prediction. For n validation forecast dates, use $\text{MAE} = n^{-1}\sum_t |Y_t^{(7)} - \widehat{Y}_t^{(7)}|$ and $\text{RMSE} = [n^{-1}\sum_t (Y_t^{(7)} - \widehat{Y}_t^{(7)})^2]^{1/2}$. A more advanced method must improve out-of-time performance and, eventually, be assessed for decision lead time and false/missed capacity warnings.

Computational pipeline. Raw event CSVs → schema and timestamp checks → daily Dog/Cat aggregation → history-only features → preliminary target and simple baselines → chronological validation → error diagnostics → manager-facing output (proposed). Completed: exploratory quality checks, daily representation, descriptive plots, preliminary labels, and baseline evaluation. Proposed: verified onsite/capacity measurements, model selection with rolling-origin folds, uncertainty estimation, and operational deployment. A validation design cannot remedy a fundamentally unmeasured target.

17. Assumptions, Limitations, and Open Questions

The approach assumes event recording is sufficiently stable for predicting future recorded flows, that recorded calendar dates are usable at day resolution, and that past patterns remain informative in future periods. The model may fail following changes in admission policies, event reporting, species mix, seasonality, shelter operations, or data extraction practices. The target does not track peak within-week occupancy; a week ending in net zero flow could still include a severe temporary pressure episode.

Highest-priority open questions: (1) Can historical onsite daily occupancy be obtained and reconciled with offsite foster animals? (2) What species/resource-specific historical capacities are available? (3) Does the end-of-day forecast arrive early enough for transfers or foster placement? (4) How will missed capacity warnings and unnecessary alerts be valued? These questions determine whether the preliminary flow task can support an operational overcrowding-risk decision. No causal benefit of interventions is claimed from these observational records.

Reproducibility. The accompanying Jupyter notebook documents data loading, preprocessing, quality diagnostics, exploratory analysis, and the 2024 baseline comparison. The GitHub repository includes the source-data snapshots, dependency requirements, and instructions for reproducing the reported results. The notebook was executed from a restarted kernel without errors in the local development environment.

18. Response to Milestone 1 Feedback

Feedback theme	Milestone 2 response	Status	Where addressed
Decision definition	Specify end-of-day decisions and a seven-day forecast horizon.	Addressed	Mathematical contract
Observational unit and timing	Separate event records from daily observations and define information available at decision time.	Addressed	DGP; mathematical contract
Overcrowding measurement	Distinguish observed net flow from unverified onsite occupancy and capacity.	Partial	Measurement and data quality; limitations
Decision value and architecture	Link forecasts to possible managerial actions and outline future alert evaluation measures.	Partial	Decision architecture; open questions

19. AI Use Statement

Generative AI (ChatGPT, OpenAI) was used to assist with code organization, preliminary analytical review, language editing, and LaTeX formatting. The numerical results are based on the Austin Animal Center records and computations documented in the accompanying notebook. The assumptions, measurement limitations, and validation design are described in this report.